

Golden Gate Assembly

Overview

Clone and assemble multiple core genetic parts of the light-controlled optogenetic circuit.

For construction of optogenetic system

A. Material

a. DNA plasmids

	Vector	Insert			
		Promoter	RBS	CDS	Terminator
Level 0	pSTK-0-sfGFP	All individual parts			
Level 1	pSTK-1A-sfGFP	Ptag	RBS	Ho1-pcyA	L3S2P21
	pSTK-1B-sfGFP	PliaG PlepA	RBS0	ccaSm3	B0015
	pSTK-1C-sfGFP	Pveg	RBS0	ccaR	B0015
	pSTK-1D-sfGFP	PcpcG2	EEBD-RBS0	Csn:Lea	B0015
				Csn:ACCD	
Level 2	pSTK-EXP-1-sfGFP	Pveg	EEBD-RBS0		B0015
					L3S2P21
	pSTK-EXP-2-sfGFP	Pveg	EEBD-RBS0	csn:AmiICP	B0015

For construction of constitute expression system

	Vector	Insert			
		Promoter	RBS	CDS	Terminator
Wildtype	pSTK-EXP1	Pveg	RBS0	SamyQ-LEA	B0015
				SamyQ-BoPep4	
				SamyQ-ACCD	
Codon optimization				SamyQ-LEA-v2t1	
				SamyQ-LEA-v2t2	
				SamyQ-BoPep4-v2t1	
				SamyQ-BoPep4-v2t2	
				SamyQ-ACCD-v2t1	
				SamyQ-ACCD-v3t1	
				Truncated	
Bopep4_G17A_1-23_NHis					
Bopep4_S15A_1-23_NHis					
Bopep4_WT_1-23_NHis					

- b. T4 DNA ligase buffer
- c. T4 DNA ligase
- d. Enzyme (dependant on which vector we use)
 - i. Bsal
 - ii. Bsal-HF
 - iii. BbsI
 - iv. BsmBI
- e. Ice bucket
- f. Sterile ddH₂O

B. Procedures

Reagent	Final Volume
Vector (10 fmol)	X µL
Inserts (20 fmol)	X µL
T4 DNA ligase	1 µL
T4 DNA ligase 10X buffer	2 µL
Enzyme	1 µL
ddH ₂ O	20 - (sum of everything else)
Total Volume	20 µL

Formula for calculating vector and insert volume:

Vector to insert ratio : 1:1, 1:2, 1:3, 1:5, 1:7

- Vector: $0.02 = \left\{ \left(\frac{X}{\text{sequence length(bp)} \times 650} \right) / 1000 / \text{DNA concentration} \right\}$

- Insert: $0.01 = \left\{ \left(\frac{X}{\text{sequence length(bp)} \times 650} \right) / 1000 / \text{DNA concentration} \right\}$

C. Run PCR program

Stages	Temperature (°C)	Time (sec)	Cycles
Initial Denaturation	95°C	2 min	1 cycle
Denaturation	95°C	30 sec	30 cycles
Annealing	65°C	40 sec	
Extension	72°C	2.5 min	
Final extension	72°C	1 min	1 cycle

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